MIL-L-52056D
16 April 1984
SUPERSEDING
MIL-L-52056C
1 November 1978

MILITARY SPECIFICATION

LIGHT, SIGNAL, SURVEYING: SELF-ILLUMINATED, WITH CARRYING CASE, FOR USE WITH RANGE POLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a target light of variable intensity and with internal changeable filters. The light shall be fitted in a carrying case.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

MMM-A-134	- Adhesive, Epoxy Resin.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6675

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:06Z
Check the source to verify that this is the current version before use.

MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polymide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13216E8100	- Light, Target, Surveying, Self-Illuminated with Carrying Case, for Use with Range Pole.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:06Z
Check the source to verify that this is the current version before use.

3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The self-illuminated surveying signal light with carrying case (hereinafter called "target light") shall be in accordance with top assembly TA13216E8100, and as specified herein.

3.1.1. Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the target light. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4).

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

3.4 Performance. The light-beam spread, sight alinement, watertightness, and level vial sensitivity of the target light shall be as specified herein and on the applicable drawings.

3.4.1 Light-beam spread. When tested as specified in 4.5.2.1, at the 4 foot interval the height of the light-beam shall be between 26 and 40 inches and the width of the light-beam shall be between 2.5 inches and 4 inches.

3.4.2 Sight alinement. When tested as specified in 4.5.2.2, the distance between the vertical centerline of the light-beam spread and the vertical line through the line of sight shall be .937 inch, ±.031 inch, at each of the six test distances.

3.4.3 Watertightness. When tested as specified in 4.5.2.3, there shall be no evidence of water in the target light or carrying case.

3.4.4 Level vial sensitivity. When tested as specified in 4.5.2.4, the sensitivity of the level vial shall be not less than 5 minutes or more than 10 minutes of arc per 2-mm displacement.

3.5 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:06Z
Check the source to verify that this is the current version before use.

conforming to MIL-V-173, composition I or II, as applicable (see 3.5.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.5.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management District.

3.6 Identification marking. The target light carrying case shall be identified in accordance with top assembly TA13216E8100.

3.7 Instruction plate. Each target light shall have an instruction plate in accordance with drawing 13226E1151. Plates shall be attached by screws, bolts, or rivets, or by adhesive conforming to MMM-A-134, epoxy resin, type I.

3.8 Treatment and painting. The portions of the target light normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704.

3.9 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation shall be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-C-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.10 Workmanship. All parts, components, and assemblies of the target light including castings, forgings, molded parts, stampings, seals, and machined surfaces shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. External surfaces shall be free from burrs, sharp edges, and corners except where sharp edges and corners are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:06Z
Check the source to verify that this is the current version before use.

requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings.

4.1.2 Disassembly inspection. Failure of any examination or test by the first produced target light(s) shall be cause for disassembly, in the presence of a Government representative, of the first produced target light(s) to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerance, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced target light(s). Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article target light shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The first article target light shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection or for performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.5 percent defective for major defects and 6.5 percent defective for minor defects.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:06Z
Check the source to verify that this is the current version before use.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested in accordance with the schedule shown in table I. AQL shall be 1.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The target light shall be examined as specified herein for the following defects:

Major

101. Dimensions not as specified.
102. Material not as specified.
103. Caps loose fitting.
104. Parts missing or not as specified.
105. Lamp filament cannot be centered in the aperture by external adjustment.
106. Circular level vial not sufficiently illuminated for reading at night.
107. Front sight not sufficiently illuminated and rear sight not sufficiently luminous for sighting at night.
108. Filters do not completely filter light-beam.
109. Locking wedge does not secure the target light firmly to range pole.
110. Rheostat does not extinguish the light in "off" position.
111. Treatment and painting or fungus-resistant treatment not as specified.
112. Knob on filter assembly does not turn freely and smoothly.
113. Mirror and rear sight assembly does not remain hinged at desired positions.
114. Dissimilar metals not treated as specified.
115. Workmanship not as specified.

Minor

201. Target light fits loosely in carrying case.
202. Identification marking and instruction plate incorrect, illegible, or missing.

4.5.2 Tests. The target light shall be tested as specified herein in accordance with the schedule established in table I.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:06Z
Check the source to verify that this is the current version before use.

TABLE I. Inspection schedule.

Test	Test paragraph	First Article target light	Requirement paragraph	Quality conformance
Light-beam spread	4.5.2.1	X	3.4.1	X
Sight alinement	4.5.2.2	X	3.4.2	X
Watertightness	4.5.2.3	X	3.4.3	X
Level vial sensitivity	4.5.2.4	X	3.4.4	X

4.5.2.1 Light-beam spread. The light-beam spread test may be performed at the same time as the sight-alinement test. Measure the height and width of the beam spread at the 4-foot interval. Nonconformance to 3.4.1 shall constitute failure of this test.

4.5.2.2 Sight alinement. The sight alinement may be determined in conjunction with the light-beam spread at the maximum intensity in the following manner: The light beam shall be projected upon a flat surface held at right angles to the axis of the beam at 1 foot increments ranging from 1 to 6 feet, inclusive. The distances shall be measured from the light filament adjusting screws to the flat surface. At each increment, the vertical centerline of the beam spread shall be measured and marked by a straight line upon the flat surface. At the same time, the line of sight is projected upon the same surface and marked and a line is drawn parallel to the first line passing through this marked point. Nonconformance to 3.4.2 shall constitute failure of this test.

4.5.2.3 Watertightness. The target light in normal operating position shall be subjected to a heavy spray of water at the rate of 1.0 inch per hour from above and at an angle of 15 to 20 degrees from the vertical for a period of not less than 15 minutes. The empty and closed carrying case shall be submerged in water to a depth of 3 feet for a period of not less than 20 minutes. The target light and carrying case shall then be examined. Nonconformance to 3.4.3 shall constitute failure of this test.

4.5.2.4 Level vial sensitivity. The target light shall be placed on a hinged beam, one end of which is moved by a micrometer screw. Each revolution of the screw shall result in a known angular movement of the beam. The beam shall be adjusted so that the circular level vial shows the light to be level. Bubble displacement shall be measured with a scale. The beam shall be moved until the bubble is displaced 2 mm. The displacement of the beam shall be read. Nonconformance to 3.4.4 shall constitute failure of this test. The level vials may be tested before assembly into the target light using the method described above.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:06Z
Check the source to verify that this is the current version before use.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects:

116. Containers not as specified for level A or level B. Each incorrect container shall be considered one defect.
117. Box not closed and sealed as specified for level A.
118. Lights packed together in quantities exceeding the gross weight or size limitation of the box for level A or level B.
119. Marking missing, illegible, incorrect, or incomplete for level A, level B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. The carrying case containing the light shall be placed in a close-fitting box conforming to PPP-B-636, W6c, style optional. The box shall be closed and sealed as specified for method V in the appendix to the box specification.

5.1.2 Commercial. The carrying case containing the light shall be preserved in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, level B, or commercial, as specified (see 6.2).

5.2.1 Level A. Lights, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantitites not to exceed the weight limitation of the box. Closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Lights, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-636, V3c, style optional, or PPP-B-640, class 2, style optional, in quantities not to exceed the gross weight or size limitation of the box specification. Closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Lights, preserved as specified in 5.1, shall be packed in accordance with ASTM D3951.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:06Z
Check the source to verify that this is the current version before use.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D3951. In addition, the weight and cube data shall be marked on the shipping container.

6. NOTES

6.1 Intended use. The target light is intended for use with a range pole in night survey operation.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Time frame required for submission of first article target light and number of target lights required (see 3.2).
d. Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 Batteries. The target light is designed to accommodate four dry cells conforming to MIL-B-18/9 (size D) in a series-parallel arrangement to provide 3 volts. Batteries shall not be furnished; however, the contractor shall supply all batteries used in the tests.

6.4 First article. When a first article inspection is required, the item will be tested, and should be a first produced item. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodian:
Army - ME

Review activity:
DLA - GS

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 6675-0759

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:06Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Belvoir R&D Center
ATTN: STRBE-DS
Ft. Belvoir, VA 22060

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:06Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-52056D	Light, Signal, Surveying: Self-Illuminated, with Carrying Case, for use with Range Pole

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:06Z
Check the source to verify that this is the current version before use.